

Rowan James

Project Experience Summary

Geotechnical & Tailings Engineer | BEng Civil (Cum Laude), Stellenbosch University | ECSA Candidate

Tailings Storage Facility – PFS Closure Design	
Location: Northern Cape, South Africa Year: 2025 (ongoing)	Sector: Mining / Tailings Engineering
Key Activities – Water quality sampling – On-site permeability testing – Closure options development and risk assessments	
Tailings Storage Facility – Monitoring & Reporting	
Location: Northern Cape, South Africa Year: 2024 (ongoing)	Sector: Mining / Tailings Engineering
Key Activities – Ongoing facility monitoring – Development of monthly, biannual, and annual reports – Site visits and client liaison	
Tailings Storage Facility – Operations, Maintenance & Surveillance (OMS) Manual	
Location: Northern Cape, South Africa Year: 2024	Sector: Mining / Tailings Engineering
Key Activities – Development of OMS manual for an active tailings storage facility	
Tailings Storage Facility – Continuation Study	
Location: Northern Cape, South Africa Year: 2024	Sector: Mining / Tailings Engineering
Key Activities – Lead author (working under Engineer of Record) – Water balance modelling – Slope stability assessment – Stage capacity assessment – Deposition and erosion modelling – Climate change assessment – GISTM compliance review – Concept closure design	

New Residue Storage Facility – Prefeasibility Study

Location: South Africa

Sector: Mining / Tailings Engineering

Year: 2024

Key Activities

- Design of seepage interception trenches and booster pump station
- Water balance modelling
- Capacity and freeboard assessment
- Closure design and cost assessment
- GISTM-compliant design standards

New Residue Storage Facility – Concept Design

Location: South Africa

Sector: Mining / Tailings Engineering

Year: 2024

Key Activities

- Evaluation of five footprint options
- Slope stability and capacity assessments
- Bill of quantities preparation

Rare Earth Metals Mine – Bankable Feasibility Study

Location: Angola

Sector: Mining / Tailings Engineering

Year: 2024

Key Activities

- Engineering support across multiple technical studies
- Design refinements for costing
- Contribution to feasibility documentation for project funding

TSF Closure Design – Erosion, CPTu & Buttress Design

Location: Ghana

Sector: Mining / Tailings Engineering

Year: 2023 – ongoing

Key Activities

- 300-year erosion modelling using WEPP
- CPTu investigation and data interpretation
- Slope stability modelling
- Preliminary buttress design

CPTu Geotechnical Investigation – Tailings Facilities

Location: South Africa

Sector: Geotechnical Engineering

Year: 2023

Key Activities

- Interpretation of CPTu data for two tailings storage facilities

- CAD cross-section development
- Technical report preparation

ROM Tip Pile Design Verification

Location: South Africa

Sector: Geotechnical Engineering

Year: 2023

Key Activities

- Finite element analysis (RS2) for pile design verification
- Load analysis under operational conditions

Curtain Drain Installation – Quality Assurance

Location: South Africa

Sector: Geotechnical Engineering / QAQC

Year: 2023

Key Activities

- On-site quality assurance for curtain drain installation
- QAQC reporting throughout construction